

MODULE – 3

Q1 Is labelled data supervised or unsupervised machine learning? Explain. Labeled data is used in Supervised Machine Learning.

Answer: In **supervised learning**, the dataset contains **input data along with correct output labels**. The machine learning model is trained using this labelled data, where it learns the relationship between inputs and outputs. During training, the model compares its predicted output with the actual label and improves its accuracy.

Example:

- Email classification: emails are labelled as *spam* or *not spam*
- Student result prediction: data labelled as *pass* or *fail*

Types of Machine Learning

Machine Learning is mainly classified into **four types**:

1. Supervised Learning

- **Data:** Labelled data
- **Working:** Model learns using known outputs
- **Tasks:** Classification and Regression
- **Examples:** Linear Regression, Decision Tree
- **Applications:** Result prediction, disease detection

2. Unsupervised Learning

- **Data:** Unlabelled data
- **Working:** Model finds patterns without correct answers
- **Tasks:** Clustering and grouping
- **Examples:** K-Means, Hierarchical clustering
- **Applications:** Customer grouping, data analysis

3. Semi-Supervised Learning

- **Data:** Small amount of labelled data and large amount of unlabelled data
- **Working:** Uses both labelled and unlabelled data for training
- **Applications:** Image recognition, speech recognition

4. Reinforcement Learning

- **Data:** No labelled data
- **Working:** Learns by trial and error using rewards and penalties
- **Applications:** Robotics, game playing, self-driving cars

Q2. Explain Types of Regression

Answer: Regression is a supervised learning technique used to find the relationship between a **dependent variable (output)** and one or more **independent variables (inputs)**. The type of regression used depends on the **nature of data, target variable, and shape of the regression curve**.

Some important types of regression are explained below:

1. Linear Regression

Linear regression is the **simplest form of regression** used when there is a **linear relationship** between dependent and independent variables.

General equation:

$$[Y = bX + C]$$

where,

Y = dependent variable, X = independent variable b = slope of the regression line,

C = intercept

Features:

- Fast and easy to implement
- Useful when data is limited
- Helps in detecting outliers

Application: House price prediction, sales forecasting

2. Logistic Regression

Logistic regression is used when the **dependent variable is binary** in nature such as **0/1, Yes/No, True/False**.

Features:

- Predicts probability of success or failure
- Uses **logit (sigmoid) function**
- Output is categorical

Application: Spam detection, disease prediction

3. Ridge Regression

Ridge regression is used when there is **multicollinearity** (high correlation between independent variables).

Features:

- Adds a penalty term to regression
- Reduces model complexity
- Prevents overfitting
- Known as **L2 regularization**

Application: High-dimensional data analysis

4. Lasso Regression

Lasso (Least Absolute Shrinkage and Selection Operator) regression is used for **feature selection and regularization**.

Features:

- Shrinks some coefficients to zero
- Automatically selects important features
- Known as **L1 regularization**

Application: Models with large number of features

5. Polynomial Regression

Polynomial regression is used when the relationship between variables is **non-linear**.

General equation:

$$Y = b_0 + b_1x + b_2x^2 + \dots + b_nx^n$$

Features:

- Fits curved lines
- Suitable for curvilinear data
- Uses least-squares method

Application: Growth rate prediction, trend analysis

6. Stepwise Regression

Stepwise regression builds a model by **adding or removing variables step by step**.

Methods:

- Forward selection
- Backward elimination
- Bidirectional elimination

Feature: Improves model performance by selecting optimal variables

7. ElasticNet Regression

ElasticNet regression is a **combination of Ridge and Lasso regression**.

Features:

- Handles highly correlated variables
- Suitable when predictors are more than observations
- Produces grouping effect

Application: Text mining, document optimization, SVM

Q3. Explain K-Nearest Neighbour (k-NN) Algorithm

Answer: The **k-Nearest Neighbour (k-NN)** algorithm is a **supervised machine learning algorithm** used for **classification and regression**. In this algorithm, the output labels of the training data are known, but the rule to map inputs to outputs is not explicitly learned. Instead, k-NN classifies a new data point based on its **similarity to the nearest training samples**.

The algorithm works on the principle that **similar data points lie close to each other**.

Working of k-NN Algorithm

1. Store all labelled training data.
2. Choose a value of **k** (number of nearest neighbours).
3. For a new data point, calculate the distance from all training points.
4. Select the **k closest neighbours**.
5. Assign the class by **majority voting** (classification) or by **average value** (regression).

Common distance measure: Euclidean distance

Choosing an Appropriate Value of k

- A **small k** may lead to noise and overfitting.
 - A **large k** reduces noise but may ignore small patterns.
 - Hence, selecting an appropriate k value is very important for good accuracy.
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Example of k-NN Algorithm

Consider students rated using **Academic Score** and **Extra-Curricular (EC) Score**.

If a new student has scores (8,2), the distance between this student and existing categories like *Outstanding*, *Sporty*, and *Academically Sound* is calculated using Euclidean distance.

The new student is classified into the group with the **minimum distance**, such as *Academically Sound*.

k-NN for Regression

k-NN can also be used for **regression problems** where the target variable is continuous.

In this case, the predicted value is the **average of the values of k nearest neighbours**.

Applications of k-NN

- Student performance classification
- Disease detection in medical data
- Pattern recognition
- Pharmaceutical industry for cancer detection

Pros of k-NN Algorithm

1. Simple and easy to understand
2. Easy to implement
3. No assumption about data distribution
4. Works well with multi-class problems
5. Can be used for both classification and regression

Cons of k-NN Algorithm

1. Prediction time is high
 2. Requires more memory
 3. Sensitive to the value of k
 4. Sensitive to feature scaling
 5. Performance decreases with high-dimensional data
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Q4 Explain Decision Trees with Suitable Example

Answer: A Decision Tree is a supervised machine learning algorithm used for classification and regression. It represents decisions and their possible outcomes in the form of a tree-like structure, similar to the “*Guess What*” game, where answers are obtained by asking a series of Yes/No questions.

Decision trees work well with both categorical and numerical data and can model non-linear relationships.

Basic Structure of a Decision Tree

- Root node: First decision node
- Decision nodes: Intermediate nodes that split further
- Leaf (terminal) nodes: Final output or prediction
- Branches: Conditions or decision rules

Principle / Steps of Building a Decision Tree

Example: Predict whether a customer will buy a product (Yes/No)

Step 1: Select the best attribute (Root Node)

Choose the attribute that best separates the data.

- Example: *Income*
This becomes the root node.

Step 2: Split the dataset

Divide data based on the selected attribute.

- Income → *High / Low*

Step 3: Select next best attribute

For each branch, choose another attribute that further separates the classes.

- Example: *Age*

Step 4: Continue splitting

Repeat the process until:

- All data belongs to one class, or
- No further useful split is possible

Step 5: Assign Leaf Nodes

Each terminal node gives a final decision.

- Example: *Buy = Yes* or *Buy = No*

Thus, a customer can be classified by following a path from root to leaf.

Types of Decision Trees

1. Classification Tree – Output is categorical
 - Example: Spam or Not Spam
 2. Regression Tree – Output is continuous
 - Example: Predict house price
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Advantages of Decision Trees

1. Easy to understand and interpret
 2. Mimics human decision making
 3. Works with numerical and categorical data
 4. Handles non-linear relationships well
 5. Requires less data preprocessing
 6. Fast prediction for new data
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Q5. Explain the four steps to create Decision Trees with suitable example for each step.

Answer: To create a decision tree, the following steps are necessary.

Step 1: Select a variable which best separate the classes. Set this variable as the root node.

Step 2: Divide each individual variable into the given classes thereby generating new nodes.

Note that decision trees are based on forwarding selection mechanism; so, after a split is created, it cannot be re-visited.

Step 3: Again, select a variable which best separates the classes.

Step 4: Repeat steps 2 and 3 for each node generated until further separation of individuals is not possible.

Q6. “ML Model is a combination of Task, Performance and Experience”, Explain with suitable an example.

Answer: Based on this definition, a **Machine Learning (ML) model** is a **combination of three main components**:

1. **Task (T)**
 2. **Experience (E)**
 3. **Performance (P)**
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1. Task (T)

Task (T) refers to the **real-world problem** that the ML model is designed to solve.

Examples of tasks:

- Classification (spam or not spam)
- Regression (predicting house prices)
- Clustering (grouping customers)
- Recognition (face or speech recognition)

The goal of the ML model is to perform this task accurately.

2. Experience (E)

Experience (E) is the **data or knowledge** from which the model learns.

- The model is trained using data
- It learns patterns, relationships, and trends
- Learning can be:
 - Supervised learning
 - Unsupervised learning
 - Reinforcement learning

As more data is provided, the model gains more experience and improves.

3. Performance (P)

Performance (P) measures **how well the model performs the given task** using the acquired experience.

Common performance measures:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

Performance metrics help evaluate and compare ML models.



Suitable Example: Email Spam Classification

Task (T):

Classify an email as **Spam** or **Not Spam**

Experience (E):

A large dataset of emails that are already labelled as spam or not spam

Performance (P):

Measured using **accuracy, precision, recall, and F1-score**

As the model is trained with more email data (experience), its ability to correctly classify emails (performance) improves.

Q7. A company decides to carry out its business operations on a rented space. If the cost of the rental space is Rs 20000 plus Rs 500 per employee per day, then compute monthly rental for space given that the company is open 5 days a week. Show a linear equation for this scenario with explanation.

Ans: Given

- Fixed rental cost = Rs 20,000
 - Variable cost = Rs 500 per employee per day
 - Company works 5 days a week
 - Assume 4 weeks in a month
- ⇒ Working days per month = $(5 \times 4 = 20)$ days

Let,

- (x) = number of employees (y) = monthly rental cost (in Rs)
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Step 1: Daily rental cost

Daily cost due to employees = $500 * x$

Step 2: Monthly employee-related cost = $500 * x * 20 = 10000x$

Step 3: Total monthly rental cost

y = Fixed cost + Variable cost

$y = 20000 + 10000x$

Linear Equation

$y = 10000x + 20000$

Explanation

- This equation is in the form $(y = mx + c)$, which represents a linear equation
- Here:
 - $(m = 10000) \rightarrow$ cost per employee per month (slope)
 - $(c = 20000) \rightarrow$ fixed monthly rent (intercept)
- The total monthly rent increases linearly with the number of employees

Q8. Explain overview of Clustering Techniques

Answer: Clustering is an **unsupervised learning technique** used to group data objects such that objects within the same cluster are more similar to each other than to those in other clusters.

Different clustering techniques have different strengths and limitations, and the choice of algorithm depends on factors such as **data size, cluster shape, presence of outliers, and density of data points**.

Based on these factors, clustering techniques are broadly classified into **three categories**:

1. Partitional Clustering

Partitional clustering divides the dataset into **non-overlapping clusters**, where each data object belongs to **only one cluster**.

The user must specify the **number of clusters (k)** in advance.

Examples: k-means, k-medoids

Features:

- Iteratively assigns data points to k clusters
- Nondeterministic in nature (different runs may give different results)

Advantages

1. Works well for **spherical-shaped clusters**
2. Scalable for large datasets

Limitations

1. Not suitable for complex cluster shapes
2. Does not work well with clusters of different densities

2. Hierarchical Clustering

Hierarchical clustering builds a **tree-like structure (dendrogram)** to represent nested clusters.

Types:

- **Agglomerative (bottom-up):** merges closest points
- **Divisive (top-down):** splits least similar points

The number of clusters is decided by **cutting the dendrogram** at a certain level.

Features:

- Deterministic process
- Same input always gives the same result

Advantages

1. Clearly shows relationships among data objects
2. Easy to interpret

Limitations

1. Computationally expensive and Sensitive to noise and outliers

3. Density-Based Clustering

Density-based clustering forms clusters based on **regions of high data density** separated by low-density regions.

Examples: DBSCAN, OPTICS

Features:

- Does not require predefined number of clusters
- Uses a distance threshold to identify cluster members

Advantages

1. Works well for **non-spherical cluster shapes**
2. Handles noise and outliers effectively

Limitations

1. Not suitable for high-dimensional data
2. Difficult to detect clusters with varying densities

Q9. Explain K-Means Algorithm.

Answer: K-Means is an **unsupervised learning clustering algorithm** used to divide a dataset into **k distinct clusters** such that data points within the same cluster are **highly similar**, while data points in different clusters are **dissimilar**. It is one of the **simplest and most widely used clustering techniques**.

K-means does not use labelled data and finds relationships among observations based on similarity.

Properties of Clusters

A good clustering result must satisfy the following properties:

1. **Intra-cluster similarity:** Data points within a cluster should be close to each other.
2. **Inter-cluster dissimilarity:** Data points belonging to different clusters should be far apart.

Example:

In a banking system, customers can be clustered based on **income** and **debt** to perform targeted marketing.

Working of K-Means Algorithm

The K-Means algorithm works in the following steps:

1. **Choose k:** Specify the number of clusters (k).
 2. **Initialize centroids:** Randomly select k data points as initial cluster centroids.
 3. **Cluster assignment:** Assign each data point to the nearest centroid using **Euclidean distance**.
 4. **Centroid update:** Recalculate the centroid of each cluster by computing the mean of all points in that cluster.
 5. **Repeat:** Repeat the assignment and update steps until the centroids no longer change (convergence).
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Evaluation Metrics for K-Means Clustering

1. Inertia

- Measures the **sum of distances of points from their cluster centroid**
- Represents **intra-cluster distance**
- Lower inertia indicates better clustering

2. Dunn Index

- Ratio of **minimum inter-cluster distance** to **maximum intra-cluster distance**
 - Higher Dunn index indicates better separation between clusters
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Applications of K-Means Clustering

1. Customer segmentation
 2. Document clustering
 3. Image segmentation
 4. Recommendation systems
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Pros of K-Means Algorithm

1. Simple and easy to implement
 2. Computationally fast
 3. Efficient for large datasets
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Cons of K-Means Algorithm

1. Number of clusters (k) must be predefined
 2. Sensitive to outliers
 3. Different initial centroids may give different results
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Q10. Explain Density-Based Spatial Clustering of Applications with Noise (DBSCAN)

Answer: DBSCAN is an **unsupervised machine learning clustering algorithm** used to group data points based on **density**. Unlike K-Means, DBSCAN does **not require the number of clusters to be predefined** and can effectively identify **noise (outliers)** in the dataset.

Why DBSCAN is needed

Traditional clustering methods like **K-Means** have some limitations:

- Every data point is forced into a cluster, even if it is an outlier.
- The number of clusters (k) must be specified in advance.
- Small changes in data can significantly change the clusters.

DBSCAN overcomes these problems by forming clusters only where data points are **densely packed** and treating sparse points as **noise**.

Basic Idea of DBSCAN

A **cluster** in DBSCAN is defined as a region with **high density of points**, separated by regions of **low density**.

DBSCAN can:

- Find clusters of **arbitrary shapes and sizes**
 - Handle **outliers**
 - Work well with **noisy real-world data**
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Important Parameters of DBSCAN

1. ϵ (Epsilon)

- It defines the **maximum distance** between two points to be considered neighbors.
- If ϵ is too small $\rightarrow$ many points remain unclustered.
- If ϵ is too large $\rightarrow$ different clusters may merge.

2. minPts

- Minimum number of points required in an ϵ -neighborhood to form a dense region.
 - Rule of thumb:
minPts $\geq D + 1$, where D is the number of dimensions.
 - For noisy or large datasets, minPts should be larger.
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Types of Points in DBSCAN

1. Core Point

A point having at least **minPts** points within distance ϵ .

2. Border Point

A point that lies within ϵ distance of a core point but does not have enough neighbors to be a core point itself.

3. Noise Point (Outlier)

A point that is neither a core nor a border point.

Concepts Used in DBSCAN

- **Density Reachability:**

A point is density-reachable if it lies within ϵ distance of a core point.

- **Density Connectivity:**

Two points are connected if there exists a chain of density-reachable points between them.

Algorithmic Steps of DBSCAN

1. Select an unvisited point randomly from the dataset.
 2. Find all points within ϵ distance.
 3. If the number of points $\geq$ minPts, form a cluster.
 4. Expand the cluster by checking neighbors recursively.
 5. If the condition is not satisfied, mark the point as noise.
 6. Repeat until all points are visited.
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Advantages of DBSCAN

- No need to specify number of clusters

- Detects noise and outliers effectively
- Finds clusters of arbitrary shapes

Disadvantages of DBSCAN

- Difficult to choose optimal ϵ value
 - Struggles with datasets having varying densities
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Q11. Naïve Bayes Algorithm

Naïve Bayes is a **probabilistic supervised machine learning algorithm** based on **Bayes Theorem**. It is widely used for **classification problems** such as spam filtering, document classification, and sentiment analysis. The term “**naïve**” indicates the assumption that all features used in the model are **independent of each other**.

Bayes Theorem (Bayes Rule): Bayes Theorem helps us find the probability of a class given the features.

Naïve Bayes Algorithm

In real-world problems, there are multiple X variables. So, when the features are independent, the Bayes Rule can be extended to what is called Naive Bayes. It is called ‘Naive’ because of the naive assumption that the X s are independent of each other. Naive Bayes classification algorithm can be used with continuous features but performs best for categorical variables.

When dealing with numeric features, the algorithm assumes that numerical values are normally distributed.

Pros and Cons of Naïve Bayes

Advantages:

1. Simple and easy to understand
2. Very fast in prediction
3. Works well with small and large datasets
4. Performs well for text classification problems

Disadvantages:

1. Assumes complete independence among features
 2. Zero probability problem (handled using Laplace correction)
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Applications of Naïve Bayes

1. **Text Classification** – Categorizing documents
2. **Spam Filtering** – Identifying spam emails
3. **Sentiment Analysis** – Positive, negative, neutral opinions
4. **Recommendation Systems** – Predicting user preferences

Q12. Define conditional probability and the Bayes rule with examples

Conditional Probability

Conditional probability is the probability of an event occurring **given that another event has already occurred**. It focuses only on the outcomes where the given condition is true.

Definition:

If A and B are two events, then the conditional probability of A given B is:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

where

- $P(A|B)$ is the probability of event A given B
- $P(A \cap B)$ is the probability that both A and B occur
- $P(B)$ is the probability of event B

Example:

Consider drawing a card from a deck of 52 cards.

- Let A = Drawing a Queen
- Let B = Drawing a Spade

There is only **one Queen of Spades** and **13 Spades**.

$$P(A|B) = 1/13$$

This means the probability of getting a Queen **given that the card is a Spade** is **1/13**.

Bayes Rule (Bayes Theorem)

Bayes Rule helps in finding the probability of an event based on prior knowledge of related events. It is widely used in **machine learning** to predict the class label based on known input features.

Formula:

$$P(A|B) = P(B|A) \cdot P(A) / P(B)$$

where

- $P(A)$ = Prior probability of A
 - $P(B|A)$ = Likelihood
 - $P(B)$ = Evidence
 - $P(A|B)$ = Posterior probability
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Example:

Suppose:

- 5% of emails are spam $\rightarrow (P(\text{Spam}) = 0.05)$
- 80% of spam emails contain the word “Offer” $\rightarrow (P(\text{Offer}|\text{Spam}) = 0.8)$
- 10% of all emails contain the word “Offer” $\rightarrow (P(\text{Offer}) = 0.1)$

Using Bayes Rule:

$$P(\text{Spam}|\text{Offer}) = (0.8 * 0.05)/0.1 = 0.4$$

Interpretation:

There is a **40% probability** that an email is spam **if it contains the word “Offer”**.

Q13. Write Basic neural network model explaining the function of each layer.

Answer: **Basic Neural Network Model**

A **Neural Network (NN)** or **Artificial Neural Network (ANN)** is a machine learning model inspired by the working of the **human brain**. It consists of a number of interconnected units called **neurons** that work together to process information and learn from examples.

A **basic neural network model** consists of **three main layers**:

1. Input Layer
2. Hidden Layer(s)
3. Output Layer

Information flows from the input layer to the output layer through the hidden layers.

**1. Input Layer**

- The **input layer** is the first layer of the neural network.
- It receives **raw input data** such as images, text, numbers, or sensor values.
- Each neuron in this layer represents **one feature** of the input data.
- The input layer does not perform any computation; it only passes the input values to the next layer.

Example:

In a student performance prediction model, inputs can be attendance, marks, and study hours.

2. Hidden Layer(s)

- The **hidden layer** lies between the input and output layers.
- A neural network can have **one or multiple hidden layers**.
- Each neuron in the hidden layer:
 - Receives weighted inputs from the previous layer
 - Adds a bias
 - Applies an **activation function**
- The hidden layer performs **feature extraction and learning** by adjusting weights during training.

Role of hidden layers:

- Learn complex patterns
 - Handle non-linear relationships
 - Improve prediction accuracy
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3. Output Layer

- The **output layer** is the final layer of the neural network.
- It produces the **final result or prediction**.
- The number of neurons in this layer depends on the problem:
 - One neuron for regression
 - Multiple neurons for classification
- An appropriate activation function is applied to generate meaningful output.

Example:

- Output = Pass / Fail
 - Output = Category of image
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Q14. Explain how Machine Learning, Deep Learning, and Natural Language Processing techniques can be applied to solve real-world problems in Artificial Intelligence.

Answer: Application of Machine Learning, Deep Learning, and Natural Language Processing in Artificial Intelligence. Artificial Intelligence (AI) uses different techniques such as Machine Learning (ML), Deep Learning (DL), and Natural Language Processing (NLP) to solve real-world problems. Each technique has its own role and application areas.

1. Machine Learning (ML)

Machine Learning is a technique in which computers learn from data and improve their performance without being explicitly programmed. ML algorithms identify patterns in data and make predictions or decisions.

Real-world applications of ML:

- Spam Email Detection: ML models classify emails as spam or not spam.
- Recommendation Systems: Used by Amazon and Netflix to recommend products and movies.
- Fraud Detection: Banks use ML to detect suspicious transactions.
- Weather Forecasting: Predicts weather conditions based on historical data.

ML is widely used because it is simple, efficient, and works well with structured data.

2. Deep Learning (DL)

Deep Learning is a subset of Machine Learning that uses Artificial Neural Networks with multiple hidden layers. It is especially useful for handling large and complex datasets.

Real-world applications of DL:

- Image Recognition: Used in face recognition and medical image analysis.
- Speech Recognition: Used in voice assistants like Google Assistant and Alexa.
- Autonomous Vehicles: Helps self-driving cars detect objects and lanes.
- Handwritten Character Recognition: Used in postal services and banking.

Deep Learning performs better than traditional ML when dealing with images, audio, and video data.

3. Natural Language Processing (NLP)

Natural Language Processing enables machines to understand, interpret, and respond to human language. NLP combines ML and linguistic rules to process text and speech.

Real-world applications of NLP:

- Chatbots and Virtual Assistants: Used in customer support systems.
- Sentiment Analysis: Analyses reviews and social media comments.
- Machine Translation: Google Translate converts text from one language to another.
- Text Classification: Used in document categorization and email filtering.

NLP allows computers to interact with humans in a natural and meaningful way.

Q15 Write a short note on SVM.

Answer: Support Vector Machine (SVM)

Support Vector Machine (SVM) is a supervised machine learning algorithm mainly used for classification, though it can also be used for regression. It classifies data by finding an optimal decision boundary (hyperplane) that separates data points of different classes with the maximum margin.

In SVM, each data point is represented in an n -dimensional space, where n is the number of features. The data points closest to the decision boundary are called support vectors, and they play a crucial role in defining the hyperplane.

The margin is the distance between the hyperplane and the nearest data points from each class. SVM aims to maximize this margin to improve classification accuracy and reduce misclassification.

For linearly separable data, SVM uses a straight line (in 2D) or plane (in 3D). When the data is non-linearly separable, SVM uses the kernel trick to transform the data into a higher-dimensional space where linear separation becomes possible. Common kernel functions include Linear, Polynomial, Gaussian (RBF), and Sigmoid kernels.

SVM also uses a cost parameter (C) to control misclassification. A large value of C tries to classify all training points correctly (risk of overfitting), while a small C allows some misclassification (better generalization).

Advantages of SVM

- Works well with high-dimensional data
- Effective for non-linear classification using kernels
- Memory efficient, as it uses only support vectors
- Performs well in text and image classification

Disadvantages of SVM

- Computationally expensive for large datasets
- Choice of kernel and parameters is critical
- Does not directly provide probability estimates